

SOLIDWORKS API (.NET VBA C#) — Reference (219/337): IPartDoc (part 2 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IPartDoc (part 2 of 4) (Type)

SOLIDWORKS API Help

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FeatureRevolveThin2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : FeatureRevolveThin2 Method (IPartDoc)

Obsolete. Superseded by [IFeatureManager::FeatureRevolveThin](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureRevolveThin2( _  
    ByVal Angle As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle2 As System.Double, _  
    ByVal RevType As System.Integer, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ReverseThinDir As System.Integer, _  
    ByVal Merge As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Angle As System.Double  
Dim ReverseDir As System.Boolean  
Dim Angle2 As System.Double  
Dim RevType As System.Integer  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ReverseThinDir As System.Integer  
Dim Merge As System.Boolean  
  
instance.FeatureRevolveThin2(Angle, ReverseDir, Angle2, RevType, Thickness1, Thickness2,  
ReverseThinDir, Merge)
```

C#

```
void FeatureRevolveThin2(  
    System.double Angle,  
    System.bool ReverseDir,  
    System.double Angle2,  
    System.int RevType,  
    System.double Thickness1,  
    System.double Thickness2,  
    System.int ReverseThinDir,  
    System.bool Merge  
)
```

C++/CLI

```
void FeatureRevolveThin2(  
& System.double Angle,  
& System.bool ReverseDir,  
& System.double Angle2,  
& System.int RevType,  
& System.double Thickness1,  
& System.double Thickness2,  
& System.int ReverseThinDir,  
& System.bool Merge  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle
ReverseDir
Angle2
RevType
Thickness1
Thickness2
ReverseThinDir
Merge

Visual Basic for Applications (VBA) Syntax

See [PartDoc::FeatureRevolveThin2](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

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FeatureRevolveThinCut Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : FeatureRevolveThinCut Method (IPartDoc)

Obsolete. Superseded by [IFeatureManager::FeatureRevolveThinCut](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureRevolveThinCut( _  
    ByVal Angle As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle2 As System.Double, _  
    ByVal RevType As System.Integer, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ReverseThinDir As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Angle As System.Double  
Dim ReverseDir As System.Boolean  
Dim Angle2 As System.Double  
Dim RevType As System.Integer  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ReverseThinDir As System.Integer
```

```
instance.FeatureRevolveThinCut(Angle, ReverseDir, Angle2, RevType, Thickness1, Thickness2,  
ReverseThinDir)
```

C#

```
void FeatureRevolveThinCut(  
    System.double Angle,  
    System.bool ReverseDir,  
    System.double Angle2,  
    System.int RevType,  
    System.double Thickness1,  
    System.double Thickness2,  
    System.int ReverseThinDir  
)
```

```
void FeatureRevolveThinCut(
& System.double Angle,
& System.bool ReverseDir,
& System.double Angle2,
& System.int RevType,
& System.double Thickness1,
& System.double Thickness2,
& System.int ReverseThinDir
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

ReverseDir

Angle2

RevType

Thickness1

Thickness2

ReverseThinDir

Visual Basic for Applications (VBA) Syntax

See [PartDoc::FeatureRevolveThinCut](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

SOLIDWORKS API Help

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FeatureXpert Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : FeatureXpert Method (IPartDoc)

Executes FeatureXpert, powered by SOLIDWORKS Intelligent Feature Technology (SWIFT), in parts.

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureXpert()
```

Visual Basic (Usage)

```
Dim instance As IPartDoc
```

```
instance.FeatureXpert()
```

C#

```
void FeatureXpert()
```

C++/CLI

```
void FeatureXpert();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::FeatureXpert](#).

Remarks

FeatureXpert works behind the scenes to resolve feature errors in [constant radius fillets](#), [neutral plane drafts](#), and [reference planes](#).

You must enable the FeatureXpert system option before using this method:

```
swApp.SetUserPreferenceToggle swUserEnableAutoFix, True
```

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IFeatureManager::DraftXpertChange Method](#)
[IFeatureManager::DraftXpertRemove Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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FirstFeature Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : FirstFeature Method (IPartDoc)

Gets the first feature in the part.

.NET Syntax

Visual Basic (Declaration)

```
Function FirstFeature() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object  
  
value = instance.FirstFeature()
```

C#

```
System.Object FirstFeature()
```

C++/CLI

```
System.Object^ FirstFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [feature](#) in the part

Visual Basic for Applications (VBA) Syntax

See [PartDoc::FirstFeature](#).

Example

[Suppress or Unsuppress Feature \(VBA\)](#)
[Traverse Subfeatures \(VBA\)](#)
[Autodimension All Sketches \(C#\)](#)
[Autodimension All Sketches \(VB.NET\)](#)
[Autodimension All Sketches \(VBA\)](#)

Remarks

This method returns features in the current order for the part, and this order changes when the part is edited.

Your application should not assume that:

- features retain the same relative or absolute position throughout the part's lifetime. For example, you should not assume that Sketch1 always appears before Sketch2.
- any feature has a specific name. Because features can be renamed, you cannot assume that the first reference plane feature is named Plane1.

When traversing the FeatureManager design tree, your application should use [IFeature::GetTypeName2](#) and [IFeature::GetSpecificFeature2](#) to identify specific features instead of relying solely on [IFeature::Name](#).

This method returns features in the model definition order, which is not the same as the order displayed in the user interface. See [ITreeControlItem](#) for details.

If a feature is suppressed, then this method can still access the feature.

To access the next feature in the FeatureManager design tree and subfeatures, use [IFeature::GetNextFeature](#) or [IFeature::IGetNextFeature](#) and [IFeature::GetFirstSubFeature](#) or [IFeature::IGetFirstSubFeature](#).

 See Also

- [IPartDoc Interface](#)
- [IPartDoc Members](#)
- [IPartDoc::IFirstFeature Method](#)
- [IComponent2::FirstFeature Method](#)
- [IModelDoc2::FirstFeature Method](#)
- [IModelDoc2::IFirstFeature Method](#)
- [IPartDoc::ReorderFeature Method](#)

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ForceRebuild Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ForceRebuild Method (IPartDoc)

Obsolete. Superseded by [IModelDoc2::ForceRebuild3](#).

 .NET Syntax

Visual Basic (Declaration)

```
Sub ForceRebuild()
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
  
instance.ForceRebuild()
```

C#

```
void ForceRebuild()
```

C++/CLI

```
void ForceRebuild();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [PartDoc::ForceRebuild](#).

 See Also

- [IPartDoc Interface](#)
- [IPartDoc Members](#)

SOLIDWORKS API Help

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GetBodies Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetBodies Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::GetBodies2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function GetBodies( _  
    ByVal BodyType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim BodyType As System.Integer  
Dim value As System.Object  
  
value = instance.GetBodies(BodyType)
```

C#

```
System.object GetBodies(  
    System.int BodyType  
)
```

C++/CLI

```
System.Object^ GetBodies(  
& System.int BodyType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetBodies](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

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GetBodies2 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetBodies2 Method (IPartDoc)

Gets the bodies in this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBodies2( _  
    ByVal BodyType As System.Integer, _  
    ByVal BVisibleOnly As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim BodyType As System.Integer  
Dim BVisibleOnly As System.Boolean  
Dim value As System.Object  
  
value = instance.GetBodies2(BodyType, BVisibleOnly)
```

C#

```
System.object GetBodies2(  
    System.int BodyType,  
    System.bool BVisibleOnly  
)
```

C++/CLI

```
System.Object^ GetBodies2(  
& System.int BodyType,  
& System.bool BVisibleOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Type of body as defined in [swBodyType_e](#)

BVisibleOnly

True gets only the visible bodies, false gets all of the bodies in the part

Return Value

Array of [bodies](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetBodies2](#).

Example

[Get Bodies \(C++\)](#)
[Traverse Bodies \(C++\)](#)
[Set Body for View \(C#\)](#)
[Set Body for Part \(VB.NET\)](#)
[Set Body for Part \(VBA\)](#)

Remarks

This method:

- Only supports solid and sheet (surface) body types.
- May vary the order in which bodies are returned.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IComponent2::GetBodies2 Method](#)
[IBodyFolder::GetBodies Method](#)
[IBodyFolder::IGetBodies Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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GetCorresponding Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetCorresponding Method (IPartDoc)

Obsolete. Superseded by [IModelDocExtension::GetCorresponding](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetCorresponding( _  
    ByVal InputObject As System.Object _  
) As System.Object
```

```
Dim instance As IPartDoc
Dim InputObject As System.Object
Dim value As System.Object

value = instance.GetCorresponding(InputObject)
```

C#

```
System.object GetCorresponding(
    System.object InputObject
)
```

C++/CLI

```
System.Object^ GetCorresponding(
    & System.Object^ InputObject
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InputObject

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetCorresponding](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

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GetCorrespondingEntity Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetCorrespondingEntity Method (IPartDoc)

Obsolete. Superseded by [IModelDocExtension::GetCorrespondingEntity](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetCorrespondingEntity( _
    ByVal Entity As System.Object _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IPartDoc
Dim Entity As System.Object
Dim value As System.Object

value = instance.GetCorrespondingEntity(Entity)
```

C#

```
System.object GetCorrespondingEntity(
    System.object Entity
)
```

C++/CLI

```
System.Object^ GetCorrespondingEntity(
    & System.Object^ Entity
)
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetCorrespondingEntity](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetEntityByName Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetEntityByName Method (IPartDoc)

Gets an entity (face, edge, vertex) by name.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntityByName( _  
    ByVal Name As System.String, _  
    ByVal EntityType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Name As System.String  
Dim EntityType As System.Integer  
Dim value As System.Object  
  
value = instance.GetEntityByName(Name, EntityType)
```

C#

```
System.object GetEntityByName(  
    System.string Name,  
    System.int EntityType  
)
```

C++/CLI

```
System.Object^ GetEntityByName(  
& System.String^ Name,  
& System.int EntityType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of entity

EntityType

Entity type as defined in [swSelectType_e](#) (see **Remarks**)

Return Value

[Entity](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetEntityByName](#).

Example

[Get Entity By Name \(VBA\)](#)

Remarks

Valid entityType types:

- swSelFACES
- swSelEDGES
- swSelVERTICES

Another way to obtain an [IEntity](#) object based on its name is to traverse the body topology (for example, use [IBody2::GetFirstFace](#) or [IBody2::IGetFirstFace](#) and so on) and use [IModelDoc2::GetEntityName](#) or [IModelDoc2::IGetEntityName](#) to determine when you have located the desired entity.

Because the names of entities are stored with the part document, this method resides on the [IPartDoc](#) object.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::SetEntityName Method](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::IDeleteEntityName Method](#)

SOLIDWORKS API Help

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GetEntityName Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetEntityName Method (IPartDoc)

Gets the name of the specified entity.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetEntityName( _  
    ByVal Entity As System.Object _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Entity As System.Object  
Dim value As System.String  
  
value = instance.GetEntityName(Entity)
```

C#

```
System.string GetEntityName(  
    System.object Entity  
)
```

C++/CLI

```
System.String^ GetEntityName(  
    & System.Object^ Entity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Entity](#)

Return Value

Name of entity

 Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetEntityName](#).

 Example

[Get Named Entities \(VBA\)](#)
[Select and Rename Object \(VBA\)](#)
[Get Cross Break Feature Data in Sheet Metal Part \(C#\)](#)
[Get Cross Break Feature Data in Sheet Metal Part \(VB.NET\)](#)
[Get Cross Break Feature Data in Sheet Metal Part \(VBA\)](#)

 Remarks

Valid for faces, edges, and vertices only.

SOLIDWORKS suggests that you use the more comprehensive method [IModelDoc2::GetEntityName](#) or [IModelDoc2::IGetEntityName](#) instead of this method.

 See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::SetEntityName Method](#)
[IPartDoc::DeleteEntityName Method](#)

SOLIDWORKS API Help

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GetExplodedViewConfigurationName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetExplodedViewConfigurationName Method (IPartDoc)

Gets the name of the configuration for the specified explode view of this multibody part.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewConfigurationName( _  
    ByVal ExplodedViewName As System.String _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ExplodedViewName As System.String  
Dim value As System.String  
  
value = instance.GetExplodedViewConfigurationName(ExplodedViewName)
```

C#

```
System.string GetExplodedViewConfigurationName(  
    System.string ExplodedViewName  
)
```

C++/CLI

```
System.String^ GetExplodedViewConfigurationName(  
& System.String^ ExplodedViewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ExplodedViewName
Name of the explode view whose configuration to get

Return Value

Name of the configuration for ExplodedViewName

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetExplodedViewConfigurationName](#).

Example

See the [IPartExplodeStep](#) example.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetExplodedViewNames Method \(\)](#)
[IPartDoc::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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GetExplodedViewCount Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetExplodedViewCount Method (IPartDoc)

Gets the number of explode views in the specified configuration of this multibody part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewCount( _  
    ByVal ConfigurationName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ConfigurationName As System.String  
Dim value As System.Integer  
  
value = instance.GetExplodedViewCount(ConfigurationName)
```

C#

```
System.int GetExplodedViewCount(  
    System.string ConfigurationName  
)
```

C++/CLI

```
System.int GetExplodedViewCount(  
& System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName
Name of the configuration

Return Value

Number of explode views in the specified configuration

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetExplodedViewCount](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::ShowExploded Method \(\)](#)
[IPartDoc::GetExplodedViewNames Method \(\)](#)
[IPartDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IPartDoc::CreateExplodedView Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

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GetExplodedViewNames Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetExplodedViewNames Method (IPartDoc)

Gets the names of all the explode views in the specified configuration of this multibody part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewNames( _  
    ByVal ConfigurationName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ConfigurationName As System.String  
Dim value As System.Object  
  
value = instance.GetExplodedViewNames(ConfigurationName)
```

C#

```
System.object GetExplodedViewNames(  
    System.string ConfigurationName  
)
```

C++/CLI

```
System.Object^ GetExplodedViewNames(  
& System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName
Name of the configuration

Return Value

Array of strings of the names of the explode views for ConfigurationName

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetExplodedViewNames](#).

Example

See the [IPartExplodeStep](#) example.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IPartDoc::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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GetMateReferenceEntity Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetMateReferenceEntity Method (IPartDoc)

Gets the mate reference entity.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMateReferenceEntity() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object  
  
value = instance.GetMateReferenceEntity()
```

C#

```
System.object GetMateReferenceEntity()
```

C++/CLI

```
System.Object^ GetMateReferenceEntity();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Mate reference entity

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetMateReferenceEntity](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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GetMaterialPropertyName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetMaterialPropertyName Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::GetMaterialPropertyName2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialPropertyName( _  
    ByRef Database As System.String _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Database As System.String  
Dim value As System.String  
  
value = instance.GetMaterialPropertyName(Database)
```

C#

```
System.string GetMaterialPropertyName(  
    out System.string Database  
)
```

C++/CLI

```
System.String^ GetMaterialPropertyName(  
& [Out] System.String^ Database  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Database

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetMaterialPropertyName](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

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GetMaterialPropertyName2 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetMaterialPropertyName2 Method (IPartDoc)

Gets the names of the material database and the material for the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialPropertyName2( _  
    ByVal ConfigName As System.String, _  
    ByRef Database As System.String _  
) As System.String
```

Visual Basic (Usage)



```

Dim instance As IPartDoc
Dim ConfigName As System.String
Dim Database As System.String
Dim value As System.String

value = instance.GetMaterialPropertyName2(ConfigName, Database)

```

C#

```

System.string GetMaterialPropertyName2(
    System.string ConfigName,
    out System.string Database
)

```

C++/CLI

```

System.String^ GetMaterialPropertyName2(
    & System.String^ ConfigName,
    & [Out] System.String^ Database
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of configuration (see **Remarks**)

Database
Name of material database

Return Value

Name of material

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetMaterialPropertyName2](#).

Example

[Get Material Database Paths of Document \(VBA\)](#)
[Get Material \(VBA\)](#)

Remarks

This method gets the name of the material database and an entry in the material database that you need for the XML lookup. Open the database as an XML stream and work with the XML API to get the data.

If...	Then...
Only the default configuration exists for the part	Use "" for ConfigName
A material was not applied to the configuration	Database name and return value are blank

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetMaterialVisualProperties Method](#)
[IPartDoc::SetMaterialPropertyName2 Method](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::MaterialIdName Property](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IPartDoc::MaterialUserName Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

GetMaterialVisualProperties Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetMaterialVisualProperties Method (IPartDoc)

Gets the material visual properties for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialVisualProperties() As MaterialVisualPropertiesData
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As MaterialVisualPropertiesData  
  
value = instance.GetMaterialVisualProperties()
```

C#

```
MaterialVisualPropertiesData GetMaterialVisualProperties()
```

C++/CLI

```
MaterialVisualPropertiesData^ GetMaterialVisualProperties();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Material visual properties data](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetMaterialVisualProperties](#).

Example

[Get and Set Material Visual Properties \(VBA\)](#)
[Get and Set Material Visual Properties \(VB.NET\)](#)
[Get and Set Material Visual Properties \(C#\)](#)

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IPartDoc::MaterialIdName Property](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::MaterialUserName Property](#)
[IPartDoc::GetMaterialPropertyName2 Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetNamedEntities Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetNamedEntities Method (IPartDoc)

Gets an array of names of named entities (face, edge, vertex, and so on) in this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNamedEntities() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object
```

```
value = instance.GetNamedEntities()
```

C#

```
System.object GetNamedEntities()
```

C++/CLI

```
System.Object^ GetNamedEntities();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of named [entities](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetNamedEntities](#).

Example

[Get Named Entities \(VBA\)](#)

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::SetEntityName Method](#)
[IPartDoc::IDeleteEntityName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetNamedEntitiesCount Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetNamedEntitiesCount Method (IPartDoc)

Gets the number of named entities (face, edge, vertex, and so on) in this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNamedEntitiesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Integer
```

```
value = instance.GetNamedEntitiesCount()
```

C#

```
System.int GetNamedEntitiesCount()
```

C++/CLI

```
System.int GetNamedEntitiesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of named entities

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetNamedEntitiesCount](#).

Example

[Get Named Entities \(C++\)](#)
[Get Named Entities \(VBA\)](#)

Remarks

Call this method before calling [IPartDoc::IGetNamedEntities](#) to determine the size of that method's array.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::SetEntityName Method](#)
[IPartDoc::DeleteEntityName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetPartBox Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetPartBox Method (IPartDoc)

Gets the box enclosing the part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPartBox( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim NoConversion As System.Boolean  
Dim value As System.Object  
  
value = instance.GetPartBox(NoConversion)
```

C#

```
System.object GetPartBox(  
    System.bool NoConversion
```

)

C++/CLI

```
System.Object^ GetPartBox(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

Convert to user units or not; true retains system units, false changes to user units

Return Value

Array of 6 doubles containing the 2 diagonal points that bound the part

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetPartBox](#).

Example

[Get Part Bounding Box \(VBA\)](#)

Remarks

The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

The X,Y,Z points returned are the lower- and upper-diagonal corners that bound the part with the box sides parallel to the X, Y and Z axes. The box dimensions returned enclose the part and usually close to the minimum possible size, but this is not guaranteed.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IComponent2::GetBox Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetProcessedBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetProcessedBody Method (IPartDoc)

Obsolete. Superseded by [IBody2::GetProcessedBody2](#) and [IPartDoc::IGetProcessedBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetProcessedBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object  
  
value = instance.GetProcessedBody()
```

C#

```
System.object GetProcessedBody()
```

C++/CLI

```
System.Object^ GetProcessedBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetProcessedBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetPropertyExtension Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetPropertyExtension Method (IPartDoc)

Gets the specified property extension on this part document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension( _  
    ByVal ID As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ID As System.Integer  
Dim value As System.Object
```

```
value = instance.GetPropertyExtension(ID)
```

C#

```
System.object GetPropertyExtension(  
    System.int ID  
)
```

C++/CLI

```
System.Object^ GetPropertyExtension(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

(Size of the array returned by [IPartDoc::AddPropertyExtension](#)) - (Position of the property extension in the array) (see **Remarks**)

Return Value

Value of the property extension

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetPropertyExtension](#).

Example

[Add and Get Property Extensions \(C#\)](#)
[Add and Get Property Extensions \(VB.NET\)](#)
[Add and Get Property Extensions \(VBA\)](#)

Remarks

IPartDoc::AddPropertyExtension adds property extensions to a last-in-first-out, 1-based array and returns the size of that array. See the examples in **Example**.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interface provide more flexibility.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::ResetPropertyExtension Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetRelatedBodies Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetRelatedBodies Method (IPartDoc)

Creates a list of all the related bodies in a part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelatedBodies() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Object

value = instance.GetRelatedBodies()
```

C#

```
System.Object GetRelatedBodies()
```

C++/CLI

```
System.Object^ GetRelatedBodies();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of related bodies

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetRelatedBodies](#).

Remarks

The list contains only those bodies associated with reference surfaces. What results is a list of bodies, which is related to the model, but it does not include the part body itself.

A reference surface feature can consist of one or more surfaces knitted together. Each reference surface feature has two [IBody2](#) objects:

- One for the front faces
- One for the back faces

Each IBody2 object has one or more faces depending on whether the reference surface feature is a set of knitted surfaces or a single underlying surface. The corresponding faces for each body pair should have normal vectors that are opposite.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::EnumBodies3 Method](#)
[IPartDoc::EnumRelatedSectionedBodies2 Method](#)
[IPartDoc::EnumRelatedBodies2 Method](#)
[IPartDoc::GetRelatedSectionedBodies Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetRelatedSectionedBodies Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetRelatedSectionedBodies Method (IPartDoc)

Gets all of the related sectioned bodies in a part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelatedSectionedBodies( _  
    ByVal ViewIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ViewIn As System.Object  
Dim value As System.Object  
  
value = instance.GetRelatedSectionedBodies(ViewIn)
```

C#

```
System.object GetRelatedSectionedBodies(  
    System.object ViewIn  
)
```

C++/CLI

```
System.Object^ GetRelatedSectionedBodies(  
& System.Object^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn
[Model view](#)

Return Value

Array of related sectioned bodies.

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetRelatedSectionedBodies](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::EnumBodies3 Method](#)
[IPartDoc::EnumRelatedBodies2 Method](#)
[IPartDoc::EnumRelatedSectionedBodies2 Method](#)
[IPartDoc::GetRelatedBodies Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

GetSectionedBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetSectionedBody Method (IPartDoc)

Gets the sectioned body seen in the specified drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSectionedBody( _  
    ByVal ViewIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ViewIn As System.Object  
Dim value As System.Object  
  
value = instance.GetSectionedBody(ViewIn)
```

C#

```
System.object GetSectionedBody(  
    System.object ViewIn  
)
```

C++/CLI

```
System.Object^ GetSectionedBody(  
& System.Object^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn
[Model view](#)

Return Value

[Sectioned body](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetSectionedBody](#).

Remarks

To determine if the desired model view is currently displaying a sectioned view, use [IModelView::GetDisplayState](#).

If you need the full body representation, use [IPartDoc::GetBodies2](#) or [IPartDoc::EnumBodies3](#), which ignores the sectioning operation.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetSectionedBody2 Method](#)

GetTessNorms Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessNorms Method (IPartDoc)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessNorms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object  
  
value = instance.GetTessNorms()
```

C#

```
System.object GetTessNorms()
```

C++/CLI

```
System.Object^ GetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of floats (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessNorms](#).

Remarks

The total size of the data is:

[9 x *sizeof(float)* x (*number of triangles*)].

The format of the returned data is:

- *float* x, y, z first point's unit normal
- *float* x, y, z second point's unit normal
- *float* x, y, z third point's unit normal

for the set of triangles for the part.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

SOLIDWORKS API Help

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GetTessTriangleCount Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriangleCount Method (IPartDoc)

Gets the number of triangles that make up the shaded picture tessellation for this part.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessTriangleCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Integer

value = instance.GetTessTriangleCount()
```

C#

```
System.int GetTessTriangleCount()
```

C++/CLI

```
System.int GetTessTriangleCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of triangles

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriangleCount](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

SOLIDWORKS API Help

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GetTessTriangles Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriangles Method (IPartDoc)

Gets the triangles that make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriangles( _
    ByVal NoConversion As System.Boolean _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim NoConversion As System.Boolean
Dim value As System.Object

value = instance.GetTessTriangles(NoConversion)
```

C#

```
System.Object GetTessTriangles(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.Object^ GetTessTriangles(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True to prohibit conversion to user units from system units, false to do so

Return Value

Array of floats (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriangles](#).

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, it is recommended that you traverse the body faces and extract the topology and geometry data to create your own faceting.

The total size of the data is:

$$\left[9 \times \text{sizeof}(\text{float}) \times (\text{number of triangles}) \right].$$

The format of the data is:

- *float* x, y, z first point in meters
- *float* x, y, z second point in meters
- *float* x, y, z third point in meters

for the set of triangles for the part.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method\(\)](#)

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GetTessTriStripEdges Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriStripEdges Method (IPartDoc)

Gets the edge ID for each of the edges in the triangles that make up the tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripEdges() As System.Object
```

```
Dim instance As IPartDoc
Dim value As System.Object

value = instance.GetTessTriStripEdges()
```

C#

```
System.Object GetTessTriStripEdges()
```

C++/CLI

```
System.Object^ GetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of longs or integers (see [Long vs. Integer](#)) containing the list of edge IDs for this part document (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriStripEdges](#).

Remarks

The format of the returned data is:

DWORD <i>FaceCount</i>	
DWORD <i>StripCount</i>	
DWORD <i>EdgeCount</i>	
ForEach <i>Face</i>	
DWORD <i>StripCount</i>	
DWORD [<i>StripLengths</i>]	
ForEach <i>Strip</i>	(where this is an array of edge IDs for each strip from 1 to (VertexPerStrip-1). Element 0 indicates if that a triangle strip exists (value = 1) or a single triangle (value = 0))
Long [<i>EdgeIds</i>]	

where:

<i>FaceCount</i> :	Number of faces on the body
<i>StripCount</i> :	Total number of strips on the body
<i>EdgeCount</i> :	Total number of vertices
<i>NumStrips</i> :	Number of strips on a particular face
<i>StripLengths</i> :	Array containing the number of vertex points on particular face strip
<i>EdgeIds</i> :	Array of edge IDs for each vertex on the particular face strip

For triangle strips, *EdgeIds* is:

If the strip has n vertices, then there are $2(n-2) + 1$ edge tags. The i -th element of the tags array (starting at $i = 1$) will be the tag of the edge between vertex $\text{floor}((i-1)/2) + 1$ and vertex $\text{floor}(i/2) + 2$, where $\text{floor}(x)$ is the integer portion of x .

For single triangles, *EdgeIds* is:

The i -th element of the tags array (starting at $i = 1$) will be the tag of the edge between vertex $i-1$ (if $i=1$ then $i=n$) and vertex i .

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

Availability

SOLIDWORKS 99 SP2, datecode 1999250

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStripNorms Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriStripNorms Method (IPartDoc)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripNorms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Object  
  
value = instance.GetTessTriStripNorms()
```

C#

```
System.object GetTessTriStripNorms()
```

C++/CLI

```
System.Object^ GetTessTriStripNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the tri strip normals (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriStripNorms](#).

Remarks

The format of the returned data is:

DWORD *FaceCount*

DWORD *StripCount*

DWORD (*VertexCount*) x 3

DWORD *NumStrips*

DWORD [*VertexPerStrip*] = array from 0 to (*Numstrips*-1).

Float [*Normals*] = array of X,Y,Z normal components for each strip from 0 to (*VertexPerStrip*-1).

where

FaceCount = Number of faces on the body.

StripCount = Total number of strips on the body.

VertexCount x 3 = Total number of vertices. Multiplied by 3 to cover X, Y, and Z.

NumStrips = Number of strips on a particular face.

VertexPerStrip = Array containing the number of vertex points on particular face strip.

NormalComp = Array of X,Y,Z normal components for each vertex on the particular face strip.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStrips Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriStrips Method (IPartDoc)

Gets the vertices that make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim NoConversion As System.Boolean  
Dim value As System.Object  
  
value = instance.GetTessTriStrips(NoConversion)
```

C#

```
System.object GetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.Object^ GetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits converting to user units from system units, false does not

Return Value

Array of floats (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriStrips](#).

Remarks

The vertices are all unique for that strip. In other words, a vertex that is shared by two tessellation triangles will only appear once in the return value.

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, it is recommended that you traverse the body faces and extract the topology and geometry data to create your own faceting.

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the level of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

This method returns an array of float values in the form of:

[*NumStrips*,

VertexPerStrip1, *VertexPerStrip2*,..., *VertexPerStripN*,

VertexPoints1[*VertexPerStrip1* elements]*3,

VertexPoints2[*VertexPerStrip2* elements]*3,...,

VertexPointsN[*VertexPerStripN* elements]*3]

where:

NumStrips = number of triangle strips on a particular face.

VertexPerStripN = number of vertex points for the Nth triangle strip.

VertexPointsN = a sub-array of vertices consisting of three float values representing the x,y,z coordinate of the vertex; the sub-array's length is defined by the *VertexPerStripN* element in the array.

Because the values *NumStrips* and the elements of *VertexPerStrip* are long values packed into single values, you must [unpack](#) them before they can be used.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStripSize Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : [GetTessTriStripSize Method \(IPartDoc\)](#)

Gets the size of the array floats required to contain the data returned when calling [IPartDoc::GetTessTriStrips](#) and [IPartDoc::IGetTessTriStrips](#).

.NET Syntax

Visual Basic (Declaration)

Function GetTessTriStripSize() As System.Integer

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Integer
```

```
value = instance.GetTessTriStripSize()
```

C#

```
System.int GetTessTriStripSize()
```

C++/CLI

```
System.int GetTessTriStripSize();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of floats returned by [IPartDoc::GetTessTriStrips](#) and [IPartDoc::IGetTessTriStrips](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriStripSize](#).

Remarks

The number is calculated as (3 + *FaceCount* + *StripCount* + 3 * *VertexCount*).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[ITessellation Interface](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)

SOLIDWORKS API Help

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GetTessTriStripTextures Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : GetTessTriStripTextures Method (IPartDoc)

Gets the texture coordinate components for each vertex on each triangle strip on each face of this part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripTextures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Object

value = instance.GetTessTriStripTextures()
```

C#

```
System.object GetTessTriStripTextures()
```

C++/CLI

```
System.Object^ GetTessTriStripTextures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the triangle strip texture coordinate components (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::GetTessTriStripTextures](#).

Remarks

The format of the returned data is:

DWORD <i>FaceCount</i>
DWORD <i>StripCount</i>
DWORD <i>VertexCount</i>
DWORD <i>FaceStripCount</i>
DWORD [] <i>VertexesPerTriStrip</i>
Double [] <i>TextureCoords</i>

where:

<i>FaceCount</i> :	Number of faces on the body
<i>StripCount</i> :	Total number of tri strips on the body
<i>VertexCount</i> :	Total number of vertexes (multiplied by 2 for U and V coordinates)
<i>FaceStripCount</i> :	Number of tri strips on each face
<i>VertexesPerTriStrip</i> :	Array from 0 to (<i>FaceStripCount</i> - 1) containing the number of vertexes on each face tri strip
<i>TextureCoords</i> :	Array from 0 to (<i>VertexesPerTriStrip</i> - 1) containing the U,V texture coordinate components for each vertex on each face tri strip

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IFace2::GetTessTextures Method \(\)](#)
[IPartDoc::GetTessNorms Method \(\)](#)
[IPartDoc::GetTessTriangleCount Method \(\)](#)
[IPartDoc::GetTessTriangles Method \(\)](#)
[IPartDoc::GetTessTriStripEdges Method \(\)](#)
[IPartDoc::GetTessTriStripNorms Method \(\)](#)
[IPartDoc::GetTessTriStrips Method \(\)](#)
[IPartDoc::GetTessTriStripSize Method \(\)](#)
[IFace2::GetTessTriStripTextures Method \(\)](#)

Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

[Send comments on this topic.](#)

IBodyObject Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IBodyObject Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::GetBodies2](#) and [IPartDoc::EnumBodies3](#).

.NET Syntax

Visual Basic (Declaration)

Function IBodyObject() As Body

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body
```

```
value = instance.IBodyObject()
```

C#

```
Body IBodyObject()
```

C++/CLI

```
Body^ IBodyObject();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IBodyObject](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IBodyObject2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IBodyObject2 Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::GetBodies2](#) and [IPartDoc::EnumBodies3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IBodyObject2() As Body2
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body2  
  
value = instance.IBodyObject2()
```

C#

```
Body2 IBodyObject2()
```

C++/CLI

```
Body2^ IBodyObject2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IBodyObject2](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateFeatureFromBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateFeatureFromBody Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::ICreateFeatureFromBody4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureFromBody( _  
    ByVal Body As Body _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Body As Body  
Dim value As Feature  
  
value = instance.ICreateFeatureFromBody(Body)
```

C#

```
Feature ICreateFeatureFromBody(  
    Body Body  
)
```

C++/CLI

```
Feature^ ICreateFeatureFromBody(  
& Body^ Body  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateFeatureFromBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateFeatureFromBody2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateFeatureFromBody2 Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::ICreateFeatureFromBody4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureFromBody2( _  
    ByVal Body As Body, _  
    ByVal MakeRefSurface As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Body As Body  
Dim MakeRefSurface As System.Boolean  
Dim value As Feature  
  
value = instance.ICreateFeatureFromBody2(Body, MakeRefSurface)
```

C#

```
Feature ICreateFeatureFromBody2(  
    Body Body,  
    System.bool MakeRefSurface  
)
```

C++/CLI

```
Feature^ ICreateFeatureFromBody2(  
& Body^ Body,  
& System.bool MakeRefSurface  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

MakeRefSurface

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateFeatureFromBody2](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateFeatureFromBody3 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateFeatureFromBody3 Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::ICreateFeatureFromBody4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureFromBody3( _  
    ByVal Body As Body, _  
    ByVal MakeRefSurface As System.Boolean, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Body As Body  
Dim MakeRefSurface As System.Boolean  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.ICreateFeatureFromBody3(Body, MakeRefSurface, Options)
```

C#

```
Feature ICreateFeatureFromBody3(  
    Body Body,  
    System.bool MakeRefSurface,  
    System.int Options  
)
```

C++/CLI

```
Feature^ ICreateFeatureFromBody3(  
& Body^ Body,
```

```
& System.bool MakeRefSurface,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

MakeRefSurface

Options

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateFeatureFromBody3](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateFeatureFromBody4 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateFeatureFromBody4 Method (IPartDoc)

Creates a new imported feature from the specified temporary body.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureFromBody4( _  
    ByVal Body As Body2, _  
    ByVal MakeRefSurface As System.Boolean, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim Body As Body2  
Dim MakeRefSurface As System.Boolean  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.ICreateFeatureFromBody4(Body, MakeRefSurface, Options)
```

C#

```
Feature ICreateFeatureFromBody4(  
    Body2 Body,  
    System.bool MakeRefSurface,  
    System.int Options  
)
```

C++/CLI

```
Feature^ ICreateFeatureFromBody4(  
& Body2^ Body,  
& System.bool MakeRefSurface,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

Temporary [body](#)

MakeRefSurface

If the body cannot be knit to a solid or if a solid body already exists in this model, then True creates of a reference surface feature

Options

Option as defined by [swCreateFeatureBodyOpts_e](#)

Return Value

Newly created imported [feature](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateFeatureFromBody4](#).

Remarks

This method:

- Is intended as the final call in a knitting operation. The body that results from your knitting efforts can be converted into an imported bod feature in the SOLIDWORKS model.
- Is not limited to [IBody2](#) objects obtained from knitting operations.
- Allows the application to decide how much additional work must be done to the model before creating the feature. The options listed in the [swCreateFeatureBodyOpts_e](#) enumeration can be OR'd together to provide additional checking or simplification of the model. If you specify [swCreateFeatureBodySimplify](#) for Options, then [swCreateFeatureBodyCheck](#) is implied.

If the application passes 0 for the options argument, then the options are disabled and the model is passed without any checking. It is up to the application to ensure that the model is a valid model.

If this method fails, the error might be caused by creating a sheet (surface) body from a multiple shells body. If so, replace this method with [IPartDoc::CreateSurfaceFeatureFromBody](#) and [IPartDoc::ICreateSurfaceFeatureFromBodyCount2](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IPartDoc::CreateFeatureFromBody3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateNewBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateNewBody Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::ICreateNewBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateNewBody() As Body
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body
```

```
value = instance.ICreateNewBody()
```

C#

```
Body ICreateNewBody()
```

C++/CLI

```
Body^ ICreateNewBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateNewBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateNewBody2 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateNewBody2 Method (IPartDoc)

Creates a new body to use for import sewing operations and returns it to the caller.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateNewBody2() As Body2
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body2
```

```
value = instance.ICreateNewBody2()
```

C#

```
Body2 ICreateNewBody2()
```

C++/CLI

```
Body2^ ICreateNewBody2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateNewBody2](#).

Example

[Create Imported Surface Body from Sketch \(C#\)](#)

Remarks

The intention is that with a handle on such a (initially empty) body, appropriate construction methods (for example, [IBody2::CreateTrimmedSurface](#)) can be called that eventually amount to a non-trivial object.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::CreateNewBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ICreateSurfaceFeatureFromBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateSurfaceFeatureFromBody Method (IPartDoc)

Creates a surface feature from a body.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSurfaceFeatureFromBody() As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Feature
```

```
value = instance.ICreateSurfaceFeatureFromBody()
```

C#

```
Feature ICreateSurfaceFeatureFromBody()
```

C++/CLI

```
Feature^ ICreateSurfaceFeatureFromBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Surface [feature](#) array

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateSurfaceFeatureFromBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::CreateSurfaceFeatureFromBody Method](#)
[IPartDoc::CreateFeatureFromBody3 Method](#)
[IPartDoc::ICreateFeatureFromBody4 Method](#)
[IPartDoc::ICreateSurfaceFeatureFromBodyCount2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision number 9.0

ICreateSurfaceFeatureFromBodyCount Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateSurfaceFeatureFromBodyCount Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::ICreateSurfaceFeatureFromBodyCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSurfaceFeatureFromBodyCount( _  
    ByVal Body As Body, _  
    ByVal Options As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```

Dim instance As IPartDoc
Dim Body As Body
Dim Options As System.Integer
Dim value As System.Integer

value = instance.ICreateSurfaceFeatureFromBodyCount(Body, Options)

```

C#

```

System.int ICreateSurfaceFeatureFromBodyCount(
    Body Body,
    System.int Options
)

```

C++/CLI

```

System.int ICreateSurfaceFeatureFromBodyCount(
    & Body^ Body,
    & System.int Options
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

Options

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateSurfaceFeatureFromBodyCount](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSurfaceFeatureFromBodyCount2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ICreateSurfaceFeatureFromBodyCount2 Method (IPartDoc)

Gets the number of surface features to create from the body.

.NET Syntax

Visual Basic (Declaration)

```

Function ICreateSurfaceFeatureFromBodyCount2( _
    ByVal Body As Body2, _
    ByVal Options As System.Integer _
) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As IPartDoc
Dim Body As Body2
Dim Options As System.Integer
Dim value As System.Integer

value = instance.ICreateSurfaceFeatureFromBodyCount2(Body, Options)

```

C#

```

System.int ICreateSurfaceFeatureFromBodyCount2(
    Body2 Body,

```

```
System.int Options
)
```

C++/CLI

```
System.int ICreateSurfaceFeatureFromBodyCount2(
& Body2^ Body,
& System.int Options
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body

Body from which to create surface features

Options

Options as defined in [swCreateFeatureBodyOptions](#)

Return Value

Number of surface features to create from the body

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ICreateSurfaceFeatureFromBodyCount2](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IPartDoc::ICreateSurfaceFeatureFromBody Method](#)

[IPartDoc::ICreateFeatureFromBody4 Method](#)

[IPartDoc::CreateFeatureFromBody3 Method](#)

[IPartDoc::CreateSurfaceFeatureFromBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteEntityName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : DeleteEntityName Method (IPartDoc)

Deletes the name associated with the specified entity.

.NET Syntax

Visual Basic (Declaration)

```
Function IDeleteEntityName( _
    ByVal Entity As Entity _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim Entity As Entity
Dim value As System.Boolean
```

```
value = instance.IDeleteEntityName(Entity)
```

C#

```
System.bool IDeleteEntityName(
    Entity Entity
)
```

C++/CLI

```
System.bool IDeleteEntityName(  
& Entity^ Entity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[Entity](#)
[Entity](#) whose name to remove

Return Value

True if the name is removed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IDeleteEntityName](#).

Remarks

Deleting an entity name can cause references to fail.

For example, if an assembly contains a mate to a part face, then a name may automatically be assigned to that face. If you delete that name, then there is no guarantee that the mate will still be valid. This method was provided because the action is available in the core product. However, you should recognize the possibility of reference failures as just described.

Entity names are kept with the part document, so this method resides on the [IPartDoc](#) object.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::ISetEntityName Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

IExportToDWG Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IExportToDWG Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::IExportToDWG2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IExportToDWG( _  
    ByVal FilePath As System.String, _  
    ByVal ModelName As System.String, _  
    ByVal Action As System.Integer, _  
    ByVal ExportToSingleFile As System.Boolean, _  
    ByRef Alignment As System.Double, _  
    ByVal IsXDirFlipped As System.Boolean, _  
    ByVal IsYDirFlipped As System.Boolean, _  
    ByVal SheetMetalOptions As System.Integer, _  
    ByVal ViewsCount As System.Integer, _  
    ByRef Views As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IPartDoc
Dim FilePath As System.String
Dim ModelName As System.String
Dim Action As System.Integer
Dim ExportToSingleFile As System.Boolean
Dim Alignment As System.Double
Dim IsXDirFlipped As System.Boolean
Dim IsYDirFlipped As System.Boolean
Dim SheetMetalOptions As System.Integer
Dim ViewsCount As System.Integer
Dim Views As System.String
Dim value As System.Boolean

value = instance.IExportToDWG(FilePath, ModelName, Action, ExportToSingleFile, Alignment,
IsXDirFlipped, IsYDirFlipped, SheetMetalOptions, ViewsCount, Views)

```

C#

```

System.bool IExportToDWG(
    System.string FilePath,
    System.string ModelName,
    System.int Action,
    System.bool ExportToSingleFile,
    ref System.double Alignment,
    System.bool IsXDirFlipped,
    System.bool IsYDirFlipped,
    System.int SheetMetalOptions,
    System.int ViewsCount,
    ref System.string Views
)

```

C++/CLI

```

System.bool IExportToDWG(
    & System.String^ FilePath,
    & System.String^ ModelName,
    & System.int Action,
    & System.bool ExportToSingleFile,
    & System.double% Alignment,
    & System.bool IsXDirFlipped,
    & System.bool IsYDirFlipped,
    & System.int SheetMetalOptions,
    & System.int ViewsCount,
    & System.String^% Views
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePath

Full path to the exported DXF/DWG file

ModelName

Full path to the active part document

Action

- 1 - export sheet metal (valid only if active document contains sheet metal)
- 2 - export selected faces or loops
- 3 - export annotation views

ExportToSingleFile

True to save as one file; false to save in separate files

Alignment

Array of twelve double values that contain information related to output alignment:

Array[0], Array[1], Array[2] - XYZ coordinates of new origin

Array[3], Array[4], Array[5] - coordinates of new x direction vector

Array[6], Array[7], Array[8] - coordinates of new y direction vector

Array[9], Array[10], Array[11] - coordinates of the vector that is normal to the selected faces or loops

The last three elements of the array are valid only if Action = 2

IsXDirFlipped

True to flip the x direction; false otherwise

IsYDirFlipped

True to flip the y direction; false otherwise
SheetMetalOptions

[Bitmask](#) that contains sheet metal export options; valid only if Action = 1:

The bitmask is {7 6 5 4 3 2 1}, where each bit is either 0 or 1 as follows:

Bit #1: 1 to export geometry; 0 to not

Bit #2: 1 to include hidden edges; 0 to not

Bit #3: 1 to export bend lines; 0 to not

Bit #4: 1 to include sketches; 0 to not

Bit #5: 1 to merge coplanar faces; 0 to not

Bit #6: 1 to export library features; 0 to not

Bit #7: 1 to export forming tools; 0 to not

Example

If you want to export flat pattern geometry, bend lines, and sketches, then Bits 1, 3, and 4 are 1, the bitmask is 0001101, and you need to set SheetMetalOptions = $2^0 + 2^2 + 2^3 = 1 + 4 + 8 = 13$.

ViewsCount

Count of annotation views to export; valid only if Action = 3

Views

Array of annotation view names to export; valid only if Action = 3

Return Value

True if export is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ExportToDWG](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IPartDoc::ExportToDWG Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IExportToDWG2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IExportToDWG2 Method (IPartDoc)

Saves various aspects of a part (sheet metal, faces, loops, and annotation views) to one or more DXF/DWG files, preserving the specified filename.

.NET Syntax

Visual Basic (Declaration)

```
Function IExportToDWG2( _  
    ByVal FilePath As System.String, _  
    ByVal ModelName As System.String, _  
    ByVal Action As System.Integer, _  
    ByVal ExportToSingleFile As System.Boolean, _  
    ByRef Alignment As System.Double, _  
    ByVal IsXDirFlipped As System.Boolean, _  
    ByVal IsYDirFlipped As System.Boolean, _  
    ByVal SheetMetalOptions As System.Integer, _  
    ByVal ViewsCount As System.Integer, _  
    ByRef Views As System.String _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim FilePath As System.String
```



```

Dim ModelName As System.String
Dim Action As System.Integer
Dim ExportToSingleFile As System.Boolean
Dim Alignment As System.Double
Dim IsXDirFlipped As System.Boolean
Dim IsYDirFlipped As System.Boolean
Dim SheetMetalOptions As System.Integer
Dim ViewsCount As System.Integer
Dim Views As System.String
Dim value As System.Boolean

```

```

value = instance.IExportToDWG2(FilePath, ModelName, Action, ExportToSingleFile, Alignment,
IsXDirFlipped, IsYDirFlipped, SheetMetalOptions, ViewsCount, Views)

```

C#

```

System.bool IExportToDWG2(
    System.string FilePath,
    System.string ModelName,
    System.int Action,
    System.bool ExportToSingleFile,
    ref System.double Alignment,
    System.bool IsXDirFlipped,
    System.bool IsYDirFlipped,
    System.int SheetMetalOptions,
    System.int ViewsCount,
    ref System.string Views
)

```

C++/CLI

```

System.bool IExportToDWG2(
    & System.String^ FilePath,
    & System.String^ ModelName,
    & System.int Action,
    & System.bool ExportToSingleFile,
    & System.double% Alignment,
    & System.bool IsXDirFlipped,
    & System.bool IsYDirFlipped,
    & System.int SheetMetalOptions,
    & System.int ViewsCount,
    & System.String^% Views
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePath

Path and filename of the exported DXF/DWG file

ModelName

Path and filename of the active part document

Action

Export action as defined in [swExportToDWG_e](#) (see **Remarks**)

ExportToSingleFile

True to save as one file; false to save as multiple files

Alignment

- In-process, unmanaged C++: Pointer to an array of 12 double values that contain information related to output alignment (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

IsXDirFlipped

True to flip the x direction; false otherwise

IsYDirFlipped

True to flip the y direction; false otherwise

SheetMetalOptions

[Bitmask](#) that contains sheet metal export options; valid only if Action = swExportToDWG_e.swExportToDWG_ExportSheetMetal (See **Remarks**)

ViewsCount

Number of annotation views to export; valid only if Action = swExportToDWG_e.swExportToDWG_ExportAnnotationViews

Views

- In-process, unmanaged C++: Pointer to an array of annotation view names to export; valid only if Action = swExportToDWG_e.swExportToDWG_ExportAnnotationViews
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if export is successful; false if not

Remarks

This method supersedes IPartDoc::IExportToDWG by:

- Adding an option to SheetMetalOptions to export a sheet metal bounding box.
- Not prepending the flat-pattern name to the FilePath filename of the exported DXF/DWG when ExportToSingleFile is false.

Alignment array is as follows:

Array[0], Array[1], Array[2] - XYZ coordinates of new origin

Array[3], Array[4], Array[5] - coordinates of new x direction vector

Array[6], Array[7], Array[8] - coordinates of new y direction vector

Array[9], Array[10], Array[11] - coordinates of the vector that is normal to the selected faces or loops

NOTE:

- Passing an array of 12 zeros results in a default orientation being used. This is equivalent to not selecting any edges to be used for axis alignment in the user interface.
- Sheet-metal part exports are automatically constrained to align the flat-pattern normal to the 2D sheet normal, thereby limiting the degrees of freedom available to the API caller.
- The last three elements of the array are valid only if Action = swExportToDWG_e.swExportToDW_ExportSelectedFacesOrLoops.

SheetMetalOptions bitmask is {12 11 10 9 8 7 6 5 4 3 2 1}, where each bit is either 0 or 1 as follows:

Bit #1: 1 to export flat-pattern geometry; 0 to not

Bit #2: 1 to include hidden edges; 0 to not

Bit #3: 1 to export bend lines; 0 to not

Bit #4: 1 to include sketches; 0 to not

Bit #5: 1 to merge coplanar faces; 0 to not

Bit #6: 1 to export library features; 0 to not

Bit #7: 1 to export forming tools; 0 to not

Bit #8: 0

Bit #9: 0

Bit #10: 0

Bit #11: 0

Bit #12: 1 to export bounding box; 0 to not

Examples:

If you want to export flat-pattern geometry, bend lines, and sketches, then Bits 1, 3, and 4 are 1, the bitmask is 0001101, and you need to set SheetMetalOptions = $2^0 + 2^2 + 2^3 = 1 + 4 + 8 = 13$.

If you want to export only the bounding box, then Bit 12 is 1, the bitmask is 100000000000, and you need to set SheetMetalOptions = $2^{11} = 2048$.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::ExportToDWG2 Method \(\)](#)
[IPartDoc::ExportFlatPatternView Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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IFeatureById Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IFeatureById Method (IPartDoc)

Gets the feature with the specified ID in the part.

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureById( _  
    ByVal ID As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim ID As System.Integer
Dim value As Feature

value = instance.IFeatureById(ID)
```

C#

```
Feature IFeatureById(
    System.int ID
)
```

C++/CLI

```
Feature^ IFeatureById(
    & System.int ID
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
ID of feature

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc.IFeatureById](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc.FeatureById Method](#)
[IPartDoc.FeatureByName Method](#)
[IPartDoc.IFeatureByName Method](#)

SOLIDWORKS API Help

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IFeatureByName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IFeatureByName Method (IPartDoc)

Gets the named feature in the part.

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureByName( _
    ByVal Name As System.String _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim Name As System.String
Dim value As Feature

value = instance.IFeatureByName(Name)
```

C#

```
Feature IFeatureByName(
    System.string Name
)
```

C++/CLI

```
Feature^ IFeatureByName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Name of feature

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IFeatureByName](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::IFeatureById Method](#)
[IPartDoc::FeatureByName Method](#)
[IPartDoc::FeatureById Method](#)

SOLIDWORKS API Help

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IFirstFeature Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IFirstFeature Method (IPartDoc)

Gets the first feature in the part.

.NET Syntax

Visual Basic (Declaration)

```
Function IFirstFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Feature  
  
value = instance.IFirstFeature()
```

C#

```
Feature IFirstFeature()
```

C++/CLI

```
Feature^ IFirstFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IFirstFeature](#).

Example

[Get Sketches \(C++\)](#)

Remarks

This method returns features in the current order for the part, and this order changes when the part is edited.

Your application should not assume that:

- features retain the same relative or absolute position throughout the part's lifetime. For example, you should not assume that Sketch1 always appears before Sketch2.
- any feature has a specific name. Because features can be renamed, you cannot assume that the first reference plane feature is named Plane1.

When traversing the FeatureManager design tree, your application should use [IFeature::GetTypeName2](#) and [IFeature::GetSpecificFeature2](#) to identify specific features instead of relying solely on [IFeature::Name](#).

This method returns features in the model definition order, which is not the same as the order displayed in the user interface. See [ITreeControlItem](#) for details.

If a feature is suppressed, then this method can still access the feature.

To access the next feature in the FeatureManager design tree and subfeatures, use [IFeature::GetNextFeature](#) or [IFeature::!GetNextFeature](#) and [IFeature::GetFirstSubFeature](#) or [IFeature::!GetFirstSubFeature](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::FirstFeature Method](#)
[IPartDoc::ReorderFeature Method](#)
[IComponent2::FirstFeature Method](#)
[IModelDoc2::FirstFeature Method](#)
[IModelDoc2::!FirstFeature Method](#)

SOLIDWORKS API Help

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IGetCorrespondingEntity Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetCorrespondingEntity Method (IPartDoc)

Obsolete. Superseded by [IModelDocExtension::GetCorrespondingEntity](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCorrespondingEntity( _  
    ByVal PEntity As Entity _  
) As Entity
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim PEntity As Entity  
Dim value As Entity  
  
value = instance.IGetCorrespondingEntity(PEntity)
```

C#

```
Entity IGetCorrespondingEntity(  
    Entity PEntity  
)
```

C++/CLI

```
Entity^ IGetCorrespondingEntity(  
& Entity^ PEntity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PEntity

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetCorrespondingEntity](#).

See Also

